

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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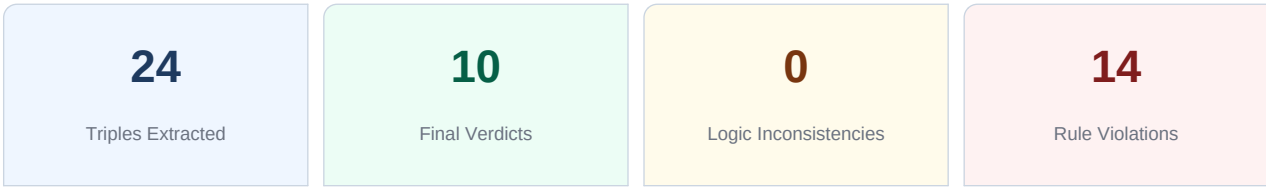
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REPORT OVERVIEW & METHODOLOGY

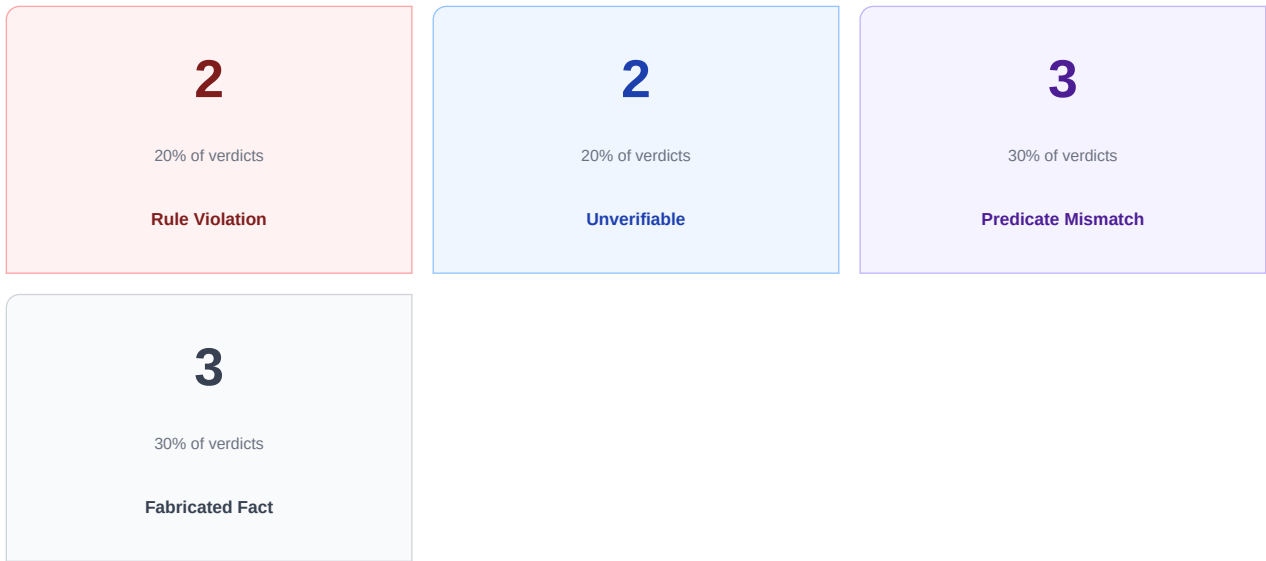
This document outlines the results of the KG-TripleGuard verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	INTEREST_RATE	30		The percentage of interest charged on a loan for a specific entity or score.
2	290	INTEREST_RATE	20		The percentage of interest charged on a loan for a specific entity or score.
3	333	INTEREST_RATE	40		The percentage of interest charged on a loan for a specific entity or score.
4	333	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
5	598	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
6	633	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
7	729	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
8	833	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
9	890	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
10	290	RISK_TIER	high prime		The category of risk assigned to a specific credit score or entity.
11	301	RISK_TIER	deep subprime		The category of risk assigned to a specific credit score or entity.
12	333	APPROVAL_ODDS	200		The likelihood or metric representing the chance of approval for a specific entity or score.
13	290	APPROVAL_ODDS	30		The likelihood or metric representing the chance of approval for a specific entity or score.
14	890	APPROVAL_ODDS	40		The likelihood or metric representing the chance of approval for a specific entity or score.
15	300	INTEREST_RATE	2		The percentage of interest charged on a loan for a specific entity or score.
16	303	INTEREST_RATE	3		The percentage of interest charged on a loan for a specific entity or score.
17	333	INTEREST_RATE	4		The percentage of interest charged on a loan for a specific entity or score.
18	777	INTEREST_RATE	6		The percentage of interest charged on a loan for a specific entity or score.
19	loan_100000_usd	REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require a certain debt-to-income ratio, collateral, or savings (e.g., 10-20% of the loan amount), but this depends on the lender's policy.		The necessary account balance or financial requirement associated with a specific loan or entity.
20	300	DESCRIPTION	Deep subprime credit score, indicating very high risk of default. Borrowers in this range face extremely high interest rates, limited loan options, and may require secured loans or co-signers.		A textual explanation or definition of the entity.

#	Subject	Predicate	Object	Type	Predicate Definition
21	300	APPROVAL_ODDS	1		The likelihood or metric representing the chance of approval for a specific entity or score.
22	305	APPROVAL_ODDS	2		The likelihood or metric representing the chance of approval for a specific entity or score.
23	405	APPROVAL_ODDS	3		The likelihood or metric representing the chance of approval for a specific entity or score.
24	730	APPROVAL_ODDS	4		The likelihood or metric representing the chance of approval for a specific entity or score.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
290	INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
333	INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
333	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
598	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
633	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
729	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
833	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
290	RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
301	RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
333	APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
290	APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
890	APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
300	INTEREST_RATE	2	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
303	INTEREST_RATE	3	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
333	INTEREST_RATE	4	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Conf.
777	INTEREST_RATE	6	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
loan_100000_usd	REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require a certain debt-to-income ratio, collateral, or savings (e.g., 10-20% of the loan amount), but this depends on the lender's policy.	REQUIRED_ACCOUNT_BALANCE	LOW
300	DESCRIPTION	Deep subprime credit score, indicating very high risk of default. Borrowers in this range face extremely high interest rates, limited loan options, and may require secured loans or co-signers.	DESCRIPTION	LOW
300	APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
305	APPROVAL_ODDS	2	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
405	APPROVAL_ODDS	3	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH
730	APPROVAL_ODDS	4	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	HIGH

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	c_interest_rate_range	VIOLATION if: subject (credit score) < 300 OR subject > 850 OR object (interest rate %) < 0 OR object > 30. A credit score of 890 VIOLATES because 890 > 850. An interest rate of -1% VIOLATES because -1 < 0. There is no exemption for out-of-range subjects — out-of-range always means violation, never auto-pass.
factually_inconsistent	RULE_VIOLATION	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	c_interest_rate_range	VIOLATION if: subject (credit score) < 300 OR subject > 850 OR object (interest rate %) < 0 OR object > 30. A credit score of 890 VIOLATES because 890 > 850. An interest rate of -1% VIOLATES because -1 < 0. There is no exemption for out-of-range subjects — out-of-range always means violation, never auto-pass.
factually_inconsistent	RULE_VIOLATION	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40	c_interest_rate_range	VIOLATION if: subject (credit score) < 300 OR subject > 850 OR object (interest rate %) < 0 OR object > 30. A credit score of 890 VIOLATES because 890 > 850. An interest rate of -1% VIOLATES because -1 < 0. There is no exemption for out-of-range subjects — out-of-range always means violation, never auto-pass.
factually_inconsistent	RULE_VIOLATION	333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	598	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	633	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	729	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	833	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	<i>c_tier_660_739</i>	VIOLATION if: (subject >= 660 OR subject <= 739) OR object (risk tier) is not semantically contains 'prime'.
factually_inconsistent	RULE_VIOLATION	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	<i>c_approval_odds_range</i>	VIOLATION if: subject (credit score) < 300 OR subject > 850 OR object (approval odds %) < 0 OR object > 100. A credit score outside 300-850 always violates. An approval odds value outside 0-100 always violates. Out-of-range subject is never an auto-pass it is always a violation.
factually_inconsistent	RULE_VIOLATION	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	<i>c_approval_odds_range</i>	VIOLATION if: subject (credit score) < 300 OR subject > 850 OR object (approval odds %) < 0 OR object > 100. A credit score outside 300-850 always violates. An approval odds value outside 0-100 always violates. Out-of-range subject is never an auto-pass it is always a violation.
factually_inconsistent	RULE_VIOLATION	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	<i>c_approval_odds_range</i>	VIOLATION if: subject (credit score) < 300 OR subject > 850 OR object (approval odds %) < 0 OR object > 100. A credit score outside 300-850 always violates. An approval odds value outside 0-100 always violates. Out-of-range subject is never an auto-pass it is always a violation.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2	Rule Violation	factually_inconsistent	RULE_VIOLATION	<i>STRICT</i>	OBJECT	1.00	The subject '300' and predicate 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE' are present in the database, but the object '2' strictly contradicts the stored value '22.94'. <i>Ev: subject: 300, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.94</i>
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	3	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	MULTIPLE	0.20	The predicate 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE' is known in the schema, but no entry for subject '303' with object '3' exists in the database.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	0.20	The predicate 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE' is known in the schema, but no entry for subject '333' with object '4' exists in the database.
777	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	6	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	0.20	The predicate 'INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE' is known in the schema, but no entry for subject '777' with object '6' exists in the database.
loan_100000_usd	REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require a certain debt-to-income ratio, collateral, or savings (e.g., 10-20% of the loan amount), but this depends on the lender's policy.	Unverifiable	unverifiable	—	NONE	—	0.00	The predicate 'REQUIRED_ACCOUNT_BALANCE' is not known in the database schema, making the triple unverifiable.
300	DESCRIPTION	Deep subprime credit score, indicating very high risk of default. Borrowers in this range face extremely high interest rates, limited loan options, and may require secured loans or co-signers.	Unverifiable	unverifiable	—	NONE	—	0.00	The predicate 'DESCRIPTION' is not known in the database schema, making the triple unverifiable.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
300	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	●●●●●●●● ●●● 1.00	The subject '300' and predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' are present in the database, but the object '1' strictly contradicts the stored value '8'. Ev: subject: 300, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 8
305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	2	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	STRICT	PREDICATE	●●●●●●●■ ■■■ 0.60	The subject '305' is present in the database, but the predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' is not a valid relation for it. Ev: subject: 305, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 305, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.61
405	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	3	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	STRICT	PREDICATE	●●●●●●●■ ■■■ 0.60	The subject '405' is present in the database, but the predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' is not a valid relation for it. Ev: subject: 405, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 405, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 17.61

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
730	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	4	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	STRICT	PREDICATE	0.60	The subject '730' is present in the database, but the predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' is not a valid relation for it. Ev: subject: 730, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: prime; subject: 730, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 7.29

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.